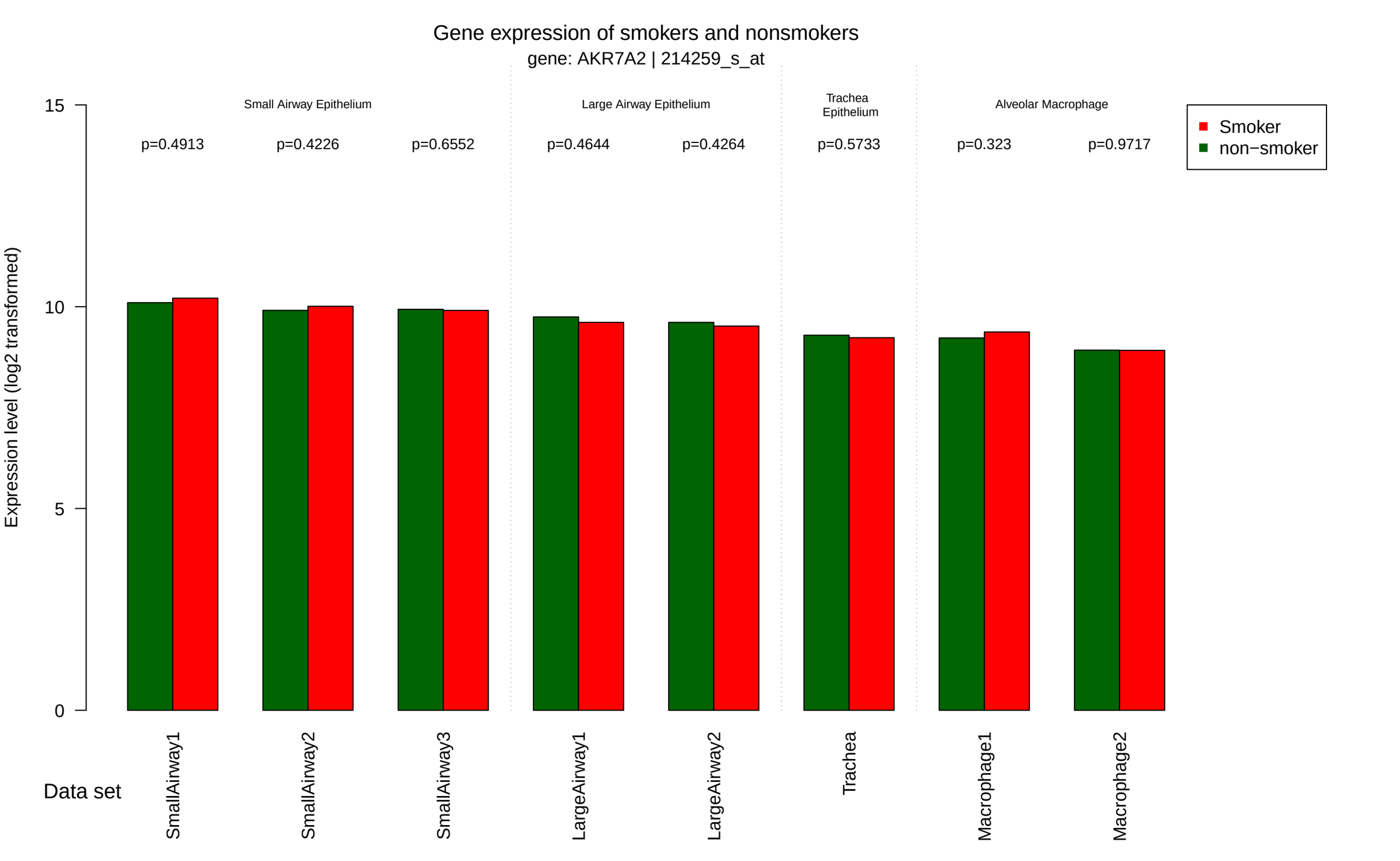
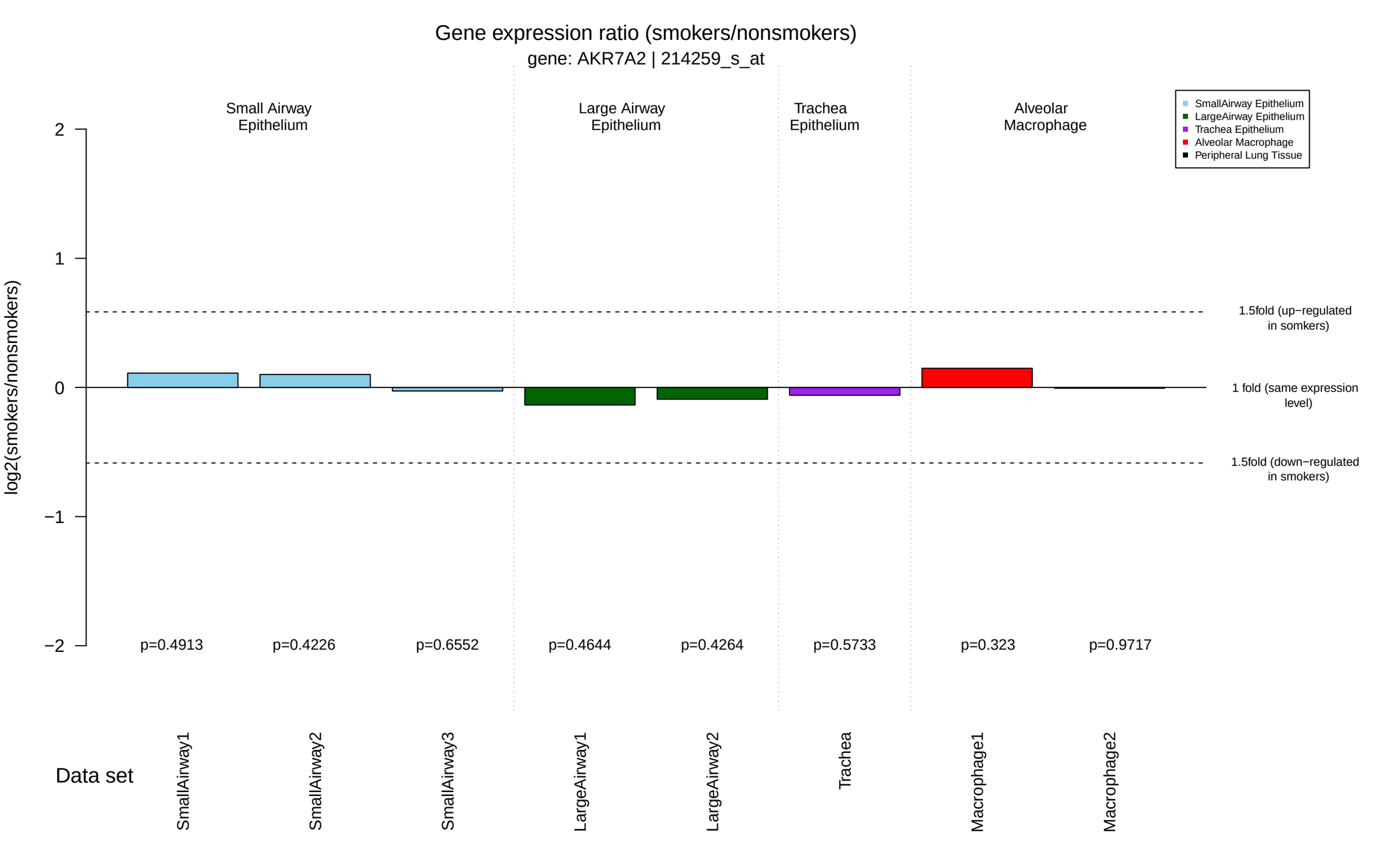
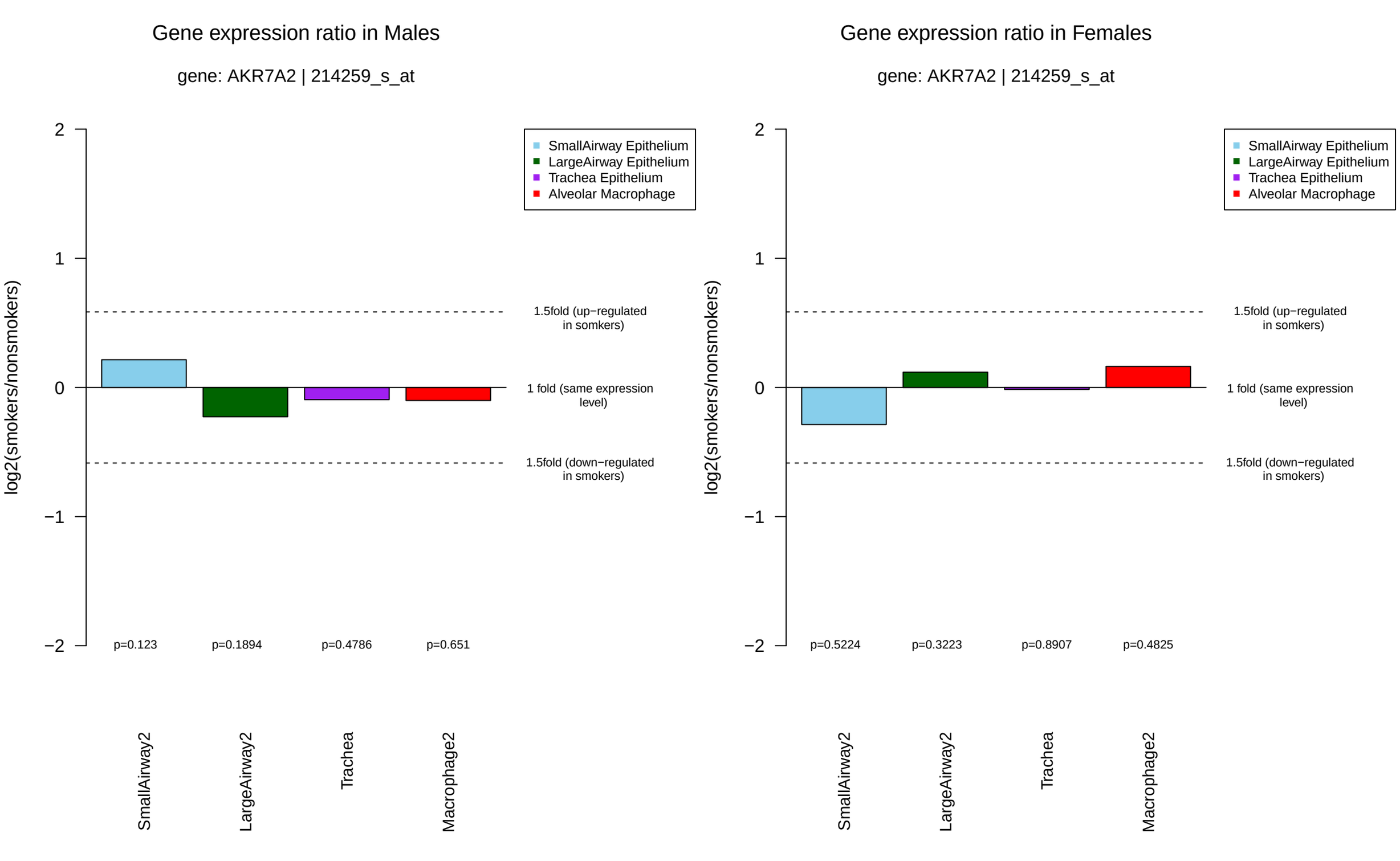
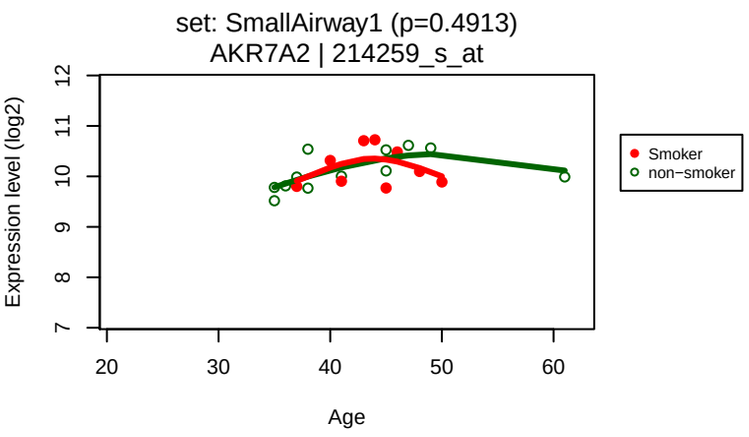
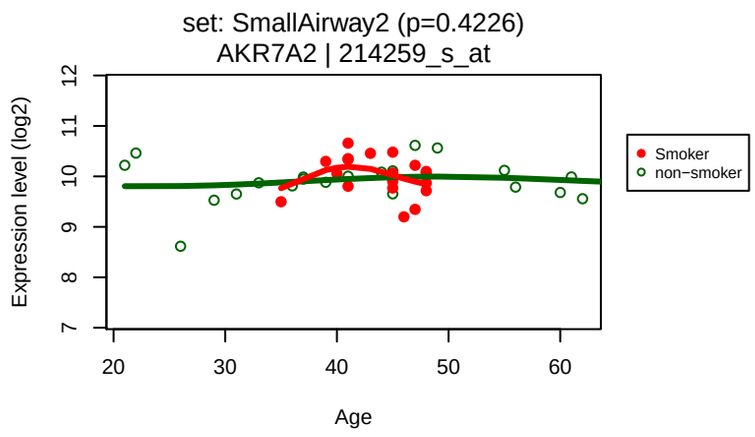
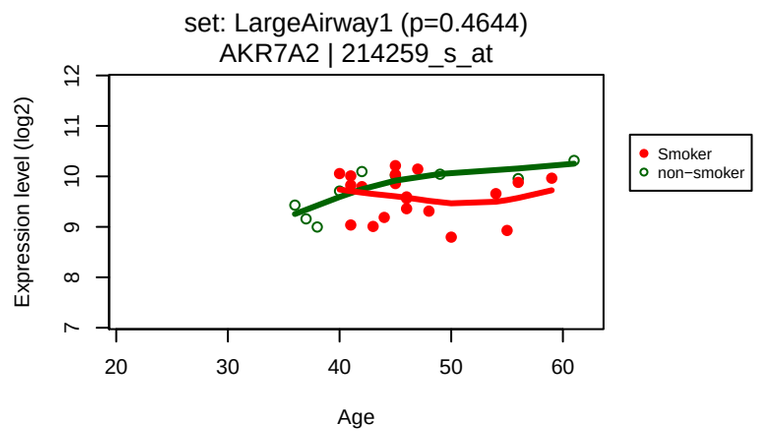

Gene expression of smokers and nonsmokers
gene: AKR7A2 | 214259_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4913
p=0.4226
p=0.6552
p=0.4644
p=0.4264
p=0.5733
p=0.323
p=0.9717
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

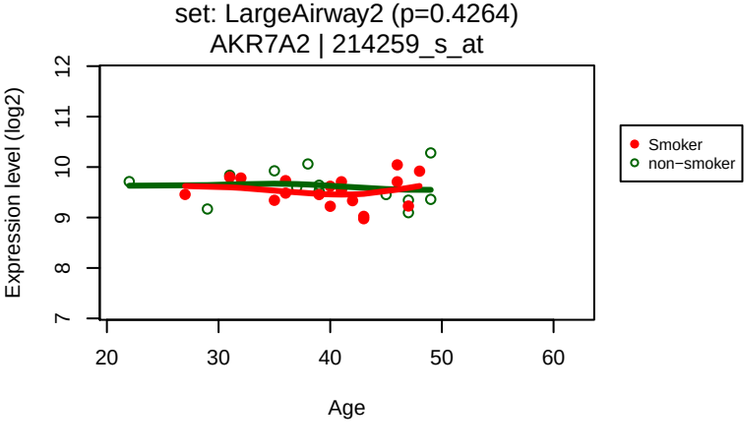

Gene expression ratio (smokers/nonsmokers)
gene: AKR7A2 | 214259_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4913
p=0.4226
p=0.6552
p=0.4644
p=0.4264
p=0.5733
p=0.323
p=0.9717
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

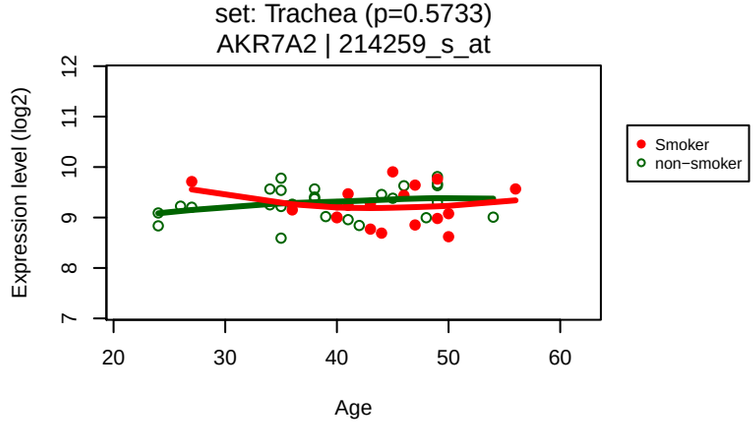

Gene expression ratio in Males
gene: AKR7A2 | 214259_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.123
p=0.1894
p=0.4786
p=0.651
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKR7A2 | 214259_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5224
p=0.3223
p=0.8907
p=0.4825
SmallAirway2
LargeAirway2
Trachea
Macrophage2

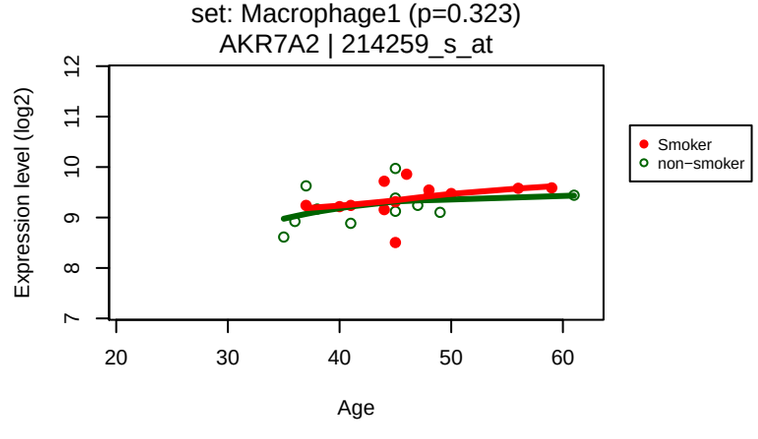

set: SmallAirway1 (p=0.4913)
AKR7A2 | 214259_s_at
Expression level (log2)
Age
Smoker
non−smoker

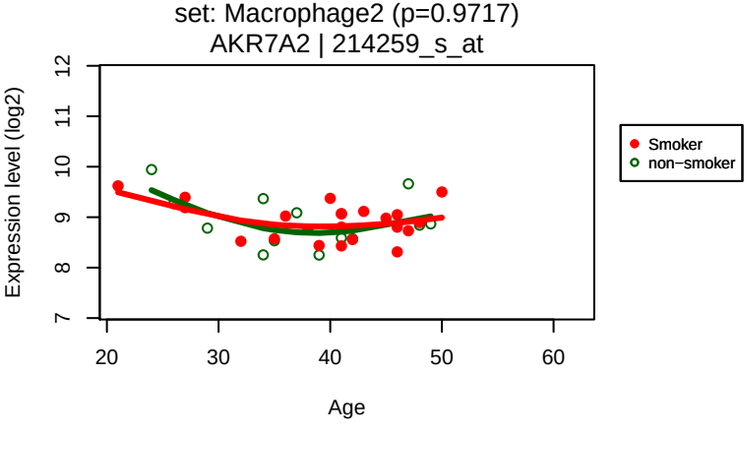

set: SmallAirway2 (p=0.4226)
AKR7A2 | 214259_s_at
Expression level (log2)
Age
Smoker
non−smoker

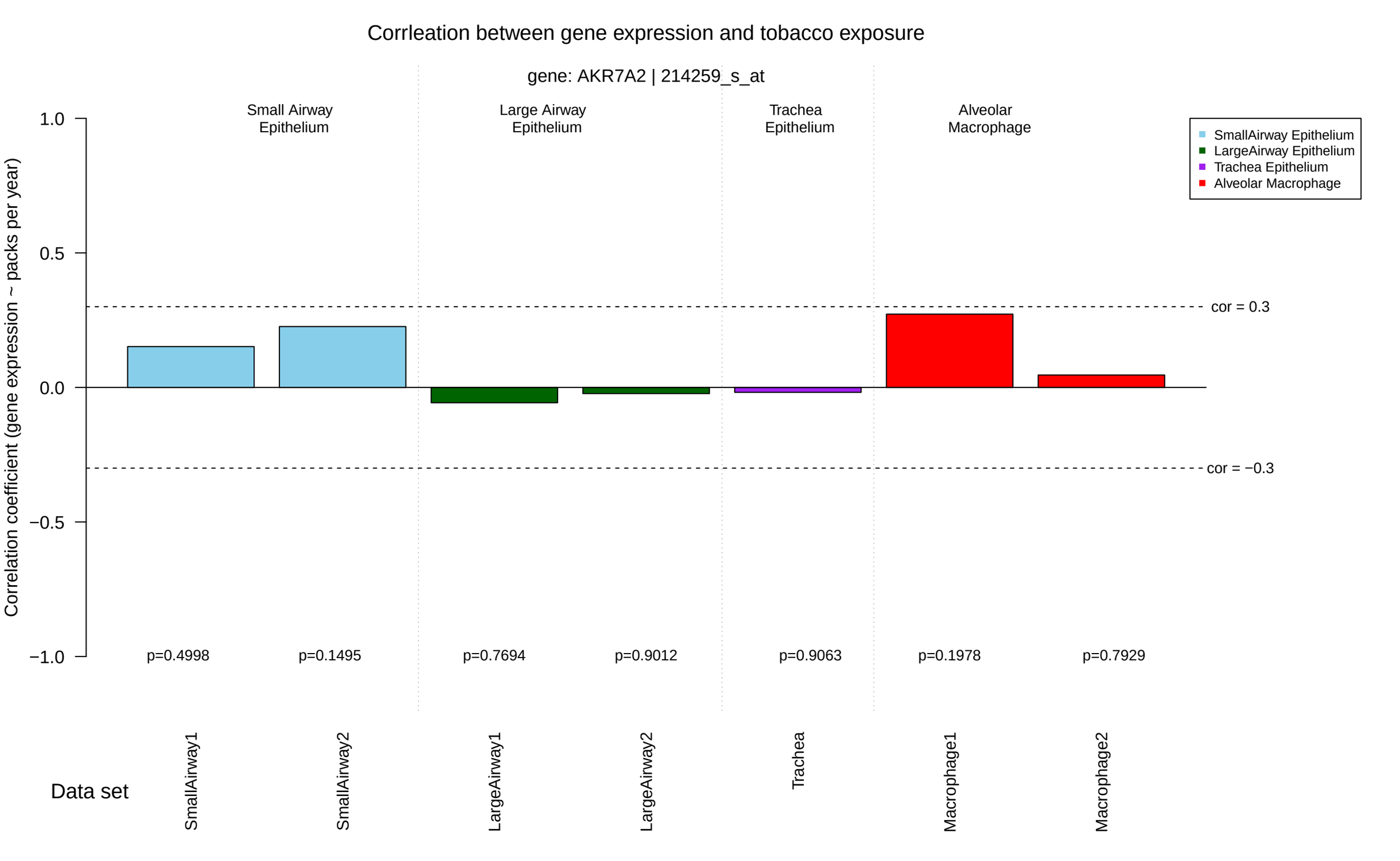

set: LargeAirway1 (p=0.4644)
AKR7A2 | 214259_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4264)
AKR7A2 | 214259_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5733)
AKR7A2 | 214259_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.323)
AKR7A2 | 214259_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9717)
AKR7A2 | 214259_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKR7A2 | 214259_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4998
p=0.1495
p=0.7694
p=0.9012
p=0.9063
p=0.1978
p=0.7929
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2